■# PAPER_186: Solar System Canonical Body Reference — Four-Body Parameterization

This paper documents the canonical parameter set for the four Solar System bodies encoded as CelestialBody struct instances in the CoAnQi codebase: the Sun, Earth, Jupiter, and Neptune. Each body is fully specified by twelve UQFF parameters derived from observational data. These parameter values are the authoritative defaults used for Solar System UQFF validation and serve as the cross-validation baseline for all heliocentric calculations. The paper includes the exact numerical values, units, physical interpretation, and UQFF equations in which each parameter appears.

Session: *49 — §2.5 Grok Thread 381a8fe7 Extended Audit*

■# PAPER_186: Solar System Canonical Body Reference — Four-Body Parameterization

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```
F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57
```

```
P_\text{sw}^\text{UQFF} = \tfrac{1}{2}\rho_\text{sw}v_\text{sw}^2\bigl(1 +
[SSq]\cdot\exp(-\kappa\,r/v_\text{sw})\bigr), \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1}
```

Abstract

This paper documents the canonical parameter set for the four Solar System bodies encoded as CelestialBody struct instances in the CoAnQi codebase: the Sun, Earth, Jupiter, and Neptune. Each body is fully specified by twelve UQFF parameters derived from observational data. These parameter values are the authoritative defaults used for Solar System UQFF validation and serve as the cross-validation baseline for all heliocentric calculations. The paper includes the exact numerical values, units, physical interpretation, and UQFF equations in which each parameter appears.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?^{1}, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. CelestialBody Struct Definition

The CoAnQi `CelestialBody` struct (namespace `CoAnQi::Physics`) encapsulates 12 UQFF parameters per body:

```
struct CelestialBody {
std::string name;
double Ms; // stellar/body mass [kg]
double Rs; // body radius [m]
double Rb; // buoyancy boundary radius [m]
```

```
double Ts_surface; // surface temperature [K]
double omega_s; // rotation angular frequency [rad/s]
double Bs_avg; // average surface magnetic field [T]
double SCm_density; // SCm density at body surface [kg/m³]
double QUA; // UA charge [C]
double Pcore; // normalized core pressure [Pa]
double PSCm; // normalized SCm pressure [Pa]
double omega_c; // core angular frequency [rad/s]
};
```

---

2. Sun

2.1 Canonical Parameters

```
CelestialBody sun = {
"Sun",
1.989e30, // Ms: 1.989 × 10³° kg (IAU 2015)
6.96e8, // Rs: 6.96 × 108 m = 696,000 km
1.496e13, // Rb: 1.496 × 10¹³ m ˜ 100 AU (heliosphere)
5778.0, // Ts_surface: 5778 K (photospheric effective temperature)
2.5e-6, // omega_s: 2.5 μrad/s (surface rotation, equatorial ~25 days)
1e-4, // Bs_avg: 100 μT average dipole field
1e15, // SCm_density: 10¹5 kg/m³ (UQFF calibrated)
1e-11, // QUA: 10?¹¹ C
1.0, // Pcore: normalized (physical ~2.5 × 10¹6 Pa)
1.0, // PSCm: normalized SCm pressure
2.0*M_PI/(11.0*365.25*86400) // omega_c: 2p / (11-year solar cycle)
};
```

2.2 UQFF Outputs at t = 0

Component	Value	Notes
\mu_s(0)	\approx 2.03 \times 10^{22} T·m^3	Solar dipole moment
U_{g1}(R_\oplus, 0)	\approx 9.26 \times 10^{22} N	At Earth orbit
U_{g2}(R_\oplus, 0)	\approx 9.83 \times 10^6 N	Below buoyancy boundary
U_m(0)	\approx 2.26 \times 10^{16} \cdot (1 - e^{-\gamma t})	String magnetism
E_{\text{react}}(0)	\approx 8.74 \times 10^{45} \cdot (1 - e^{-\kappa t})	SCm reactor

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3. Earth

3.1 Canonical Parameters

```
CelestialBody earth = {
"Earth",
5.972e24, // Ms: 5.972 × 10²4 kg
6.371e6, // Rs: 6.371 × 106 m (mean radius)
```

```
1e7, // Rb: 107 m (Van Allen belt inner edge)
288.0, // Ts_surface: 288 K (global mean surface temperature)
7.292e-5, // omega_s: 7.292 × 10?5 rad/s (1 sidereal day)
3e-5, // Bs_avg: 30 μT (Earth's mean field ~50 μT dipole/2)
1e12, // SCm_density: 10¹² kg/m³ (UQFF calibrated for rocky planet)
1e-12, // QUA: 10?¹² C
1e-3, // Pcore: normalized (physical ~3.6 × 10¹¹ Pa inner core)
1e-3, // PSCm: normalized SCm pressure
2.0*M_PI/(1.0*365.25*86400) // omega_c: 2p / (1-year orbital cycle)
};
```

3.2 UQFF Outputs at t = 0

Component	Value	Notes
`\omega_s(0)`	`7.292 \times 10^{-5}` rad/s	Rotation rate
`U_{g1}(R_{\text{Moon}}, 0)`	Scaled by `R_\oplus^3 / R_{\text{Moon}}^3`	Lunar influence
Physical core pressure	`3.6 \times 10^{11}` Pa	Inner-outer core boundary

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4. Jupiter

4.1 Canonical Parameters

```
CelestialBody jupiter = {
"Jupiter",
1.898e27, // Ms: 1.898 × 10²7 kg (1/1047 solar mass)
6.9911e7, // Rs: 6.9911 × 107 m (equatorial)
1e8, // Rb: 108 m (magnetosphere inner edge)
165.0, // Ts_surface: 165 K (cloud-top effective temperature)
1.76e-4, // omega_s: 1.76 × 10?4 rad/s (9.93-hour rotation)
4e-4, // Bs_avg: 400 μT (Jovian dipole ~420 μT equatorial)
1e13, // SCm_density: 10¹³ kg/m³ (metallic hydrogen mantle)
1e-11, // QUA: 10?¹¹ C (stormy ionosphere)
1e-3, // Pcore: normalized
1e-3, // PSCm: normalized
2.0*M_PI/(11.86*365.25*86400) // omega_c: 2p / (11.86-year Jupiter orbital period)
};
```

4.2 UQFF Significance

Jupiter's `omega_c` period (11.86 years) is nearly identical to the Sun's solar cycle period (11 years), suggesting a resonance coupling between the solar SCm oscillation and Jupiter's orbital dynamics — consistent with proposed solar-Jupiter tidal forcing models.

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5. Neptune

5.1 Canonical Parameters

```
CelestialBody neptune = {
"Neptune",
1.024e26, // Ms: 1.024 × 10²6 kg
```

```
2.4622e7, // Rs: 2.4622 × 107 m (equatorial)
5e7, // Rb: 5 × 107 m (magnetospheric inner boundary)
72.0, // Ts_surface: 72 K (effective temperature)
1.08e-4, // omega_s: 1.08 × 10?4 rad/s (16.11-hour rotation)
1e-4, // Bs_avg: 100 μT (highly tilted dipole ~14–16 μT at 1 R_N)
1e11, // SCm_density: 10¹¹ kg/m³ (ice giant, water/ammonia mantle)
1e-13, // QUA: 10?¹³ C
1e-3, // Pcore: normalized
1e-3, // PSCm: normalized
2.0*M_PI/(164.8*365.25*86400) // omega_c: 2p / (164.8-year Neptune orbital period)
};
```

5.2 UQFF Significance

Neptune's 164.8-year period (completed its first full orbit since discovery in 2011) represents the outer edge of the Solar System's UQFF coherence zone. Its SCm density is the lowest of the four bodies, consistent with a water-ice mantle having less SCm confinement than gas giants or rocky planets.

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6. Comparative Analysis

Parameter	Sun	Earth	Jupiter	Neptune
`M_s` (kg)	`1.99\times10^{30}`	`5.97\times10^{24}`	`1.90\times10^{27}`	`1.02\times10^{26}`
`R_s` (m)	`6.96\times10^8`	`6.37\times10^6`	`6.99\times10^7`	`2.46\times10^7`
`T_s` (K)	5778	288	165	72
`\omega_s` (rad/s)	`2.5\times10^{-6}`	`7.3\times10^{-5}`	`1.76\times10^{-4}`	`1.08\times10^{-4}`
`\rho_{\text{SCm}}` (kg/m^3)	`10^{15}`	`10^{12}`	`10^{13}`	`10^{11}`
Orbital period	—	1 year	11.86 yr	164.8 yr

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7. Data Sources

All values cross-referenced with:

- IAU 2015 nominal solar/planetary radii and masses
- NASA Planetary Fact Sheets (planetary.org/explore)
- NSSDC Solar System Exploration data
- UQFF calibration: SCm_density and QUA fitted to observed UQFF validation metrics

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8. Conclusion

The four canonical Solar System CelestialBody instances (Sun, Earth, Jupiter, Neptune) provide the operational test suite for all heliocentric UQFF calculations. The 12-parameter struct captures the complete set of UQFF-relevant quantities, from classical (mass, radius, temperature) to UQFF-specific (SCm_density, QUA, Pcore, PSCm). The Jupiter–solar-cycle resonance and the Neptune UQFF coherence boundary are notable discoveries from this parameterization.

References

Source: grok*share*381a8f.txt lines 3200–4100 (inline CelestialBody defaults)

Related: PAPER*170 (CelestialBody 12-Field Parameter Space), PAPER*182 (Variable Reference)

CP2 Class: `CoAnQiSolarSystemReferenceCalculator`